

1) Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
    int i= 1;
    int s =1;
    while(s <= n)
    {
        i++;
        s += i;
    }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer:(penalty regime: 0 %)

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n;
```

```
    int count = 0;
```

```
    scanf("%d", &n);
```

```
    int i = 1;
```

```
    count++;
```

```
    int s = 1;
```

```
    count++;
```

```
    while (s <= n)
```

```
    {
```

```
        count++;
```

```
        i++;
```

```
        count++;
```

```
        s = s + i;
```

```

        count++;
    }

    count++;

    printf("%d", count);

    return 0;
}

```

	Input	Expected	Got	
	9	12	12	
	4	9	9	

Passed all tests!

Correct

Marks for this submission: 1.00/1.00

2) Convert the following algorithm into a program and find its time complexity using the counter method.

```

void func(int n)
{
    if(n==1)
    {
        printf("*");
    }
    else
    {
        for(int i=1; i<=n; i++)
        {
            for(int j=1; j<=n; j++)
            {
                printf("*");
                printf("*");
                break;
            }
        }
    }
}

```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n;
```

```

int count = 1;

scanf("%d", &n);

if (n == 1)
{
    count++;
}
else
{

    for (int i = 1; i <= n; i++)
    {
        count ++;

        for (int j = 1; j <= n; j++)
        {
            count++;

            count++;

            count++;

            break;
        }
        count++;
    }
    count++;
}

printf("%d", count);

return 0;
}

```

Input	Expected	Got	
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Input	Expected	Got	
	2	12	12
	1000	5002	5002
	143	717	717

Passed all tests!

3) Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
    for (i = 1; i <= num; ++i)
    {
        if (num % i == 0)
        {
            printf("%d ", i);
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n, i;
```

```
    int count = 0;
```

```
    scanf("%d", &n);
```

```
    for (i = 1; i <= n; ++i)
```

```
    {
```

```
        count++;
```

```
        count++;
```

```
        if (n % i == 0)
```

```
        {
```

```

        count++;
    }
}

count++;

printf("%d", count);

return 0;
}

```

Input	Expected	Got	
	12	31	31
	25	54	54
	4	12	12

Passed all tests!

Correct

Marks for this submission: 1.00/1.00.

4) Convert the following algorithm into a program and find its time complexity using counter method.

```

void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}

```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n, i, j, k;
```

```
    int c = 2;
```

```

scanf("%d", &n);

for (i = n / 2; i < n; i++)
{
    c += 2;

    for (j = 1; j < n; j = 2 * j)
    {
        c += 2;

        for (k = 1; k < n; k = k * 2)
        {
            c += 2;
        }
    }
}

printf("%d", c);

return 0;
}

```

Input	Expected	Got	
4	30	30	
10	212	212	

Passed all tests!

Correct

Marks for this submission: 1.00/1.00.

5) Convert the following algorithm into a program and find its time complexity using counter method.

```

void reverse(int n)
{
    int rev = 0, remainder;
    while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n /= 10;
    }
    print(rev);
}

```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int n;
```

```
    int c = 0;
```

```
    scanf("%d", &n);
```

```
    int rev = 0, remainder;
```

```
    c++;
```

```
    while (n != 0)
```

```
    {
```

```
        c++;
```

```
        remainder = n % 10;
```

```
        c++;
```

```
        rev = rev * 10 + remainder;
```

```
        c++;
```

```
        n /= 10;
```

```
        c++;
```

```
    }
```

```
    c++;
```

```
    c++;
```

```
    printf("%d", c);
```

```
    return 0;
```

```
}
```

Input	Expected	Got	
	12	11	11
	1234	19	19

Passed all tests!

Correct

Marks for this submission: 1.00/1.00.